

Richard Feng

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Computer Science and Data Science graduate from UW Madison with hands-on experience building full stack applications, machine learning and LLM/RAG integrated systems, and data analytics pipelines. Portfolio website: richardfeng.dev.

EDUCATION

University of Wisconsin Madison

2022 – 2026

B.S. Computer Sciences, B.S. Data Science, Certificate in Mathematics | GPA: 3.762 / 4.000

EXPERIENCE

Seeds of Success, Software Engineer

January 2026 – Current

- Built full stack research document retrieval features using OpenSearch, including REST API endpoints and a Jupyter notebook integration, resulting in efficient data access for policymakers and researchers
- Reduced LLM token usage and improved response quality by developing internal context management systems to maintain live organizational knowledge
- Coordinated directly with Invisible Children's (KARA) lead developer as the primary technical point of contact to align on architecture decisions, resulting in accelerated full stack feature delivery across both teams
- Ensured a scalable, maintainable codebase by collaborating on frontend and backend architecture decisions, including component structure and routing

Senra Systems, Computer Science Capstone

February 2026 – April 2026

- Designed and built a tool to extract an iterable graph of connectors from visual schematics using RAG based analysis, resulting in automated initial workflows for wire harness manufacturing design review
- Supported persistent data storage across the application by architecting and implementing a PostgreSQL database, including schema design and integration with backend services and RAG data retrieval
- Delivered full stack features end to end, including user authentication and persistent project saving
- Cut a 3 day manual process to roughly 30 minutes at about \$0.25 per harness table and \$5 per Engineering Master, achieving 40% part spec accuracy

PROJECTS

OCT Retinal Disease Classifier

April 2026 – May 2026

- Developed a deep learning pipeline to classify 109,000+ OCT images across 4 retinal disease categories, resulting in 95% classification accuracy
- Improved DRUSEN detection precision by 6% by designing a preprocessing pipeline with multi scale Gaussian denoising and CLAHE contrast enhancement
- Achieved clinical grade performance on a 5:1 class imbalance by implementing class weighted loss functions
- Authored a research paper analyzing architecture depth vs. preprocessing trade offs; published code and documentation on GitHub

Computer Networks: Virtual Switch, Router & TCP

October 2025 – December 2025

- Implemented TCP Tahoe over UDP, including slow start, congestion avoidance, and fast retransmit, resulting in a fully functional reliable transport protocol layered entirely on raw UDP sockets
- Engineered a learning Ethernet switch in Java using Mininet and POX with MAC based forwarding and timeout aging
- Constructed a virtual IP router with longest prefix match route lookups, IPv4 checksum verification, TTL decrement, and ARP caching
- Extended the router with RIPv2 distance vector routing, including heartbeats, triggered responses, and route expiry

Asteroids — Web Based Rogue-like Game

February 2025 – April 2025

- Defined the full tech stack (React, Vite, Spring Boot, JDBC, MySQL, REST API) by prototyping login and registration demo code, resulting in a 6 person team adopting it as the project's walking skeleton architecture
- Created the gameplay rendering interface in React by porting backend game logic to the frontend, integrating real time game state via REST APIs
- Restored accurate leaderboard tracking by diagnosing and resolving a persistent score upload bug tied to session state across login/logout cycles, in collaboration with backend teammates
- Aligned team wide architecture understanding ahead of the final sprint by leading sprint planning, deliverable scoping, and team wide code review sessions for a 6 person team

TECHNICAL SKILLS & TOOLS

Languages: Python, Java, JavaScript, SQL, HTML & CSS, R

Machine Learning & AI: PyTorch, TensorFlow, LLM/RAG Integration, Deep Learning, Data Pipelines

Frameworks & Libraries: React, Next.js, Node.js, Spring Boot, REST APIs

Tools & Databases: Git, GitLab, Docker, Jupyter, Maven, MySQL, PostgreSQL, OpenSearch, LDAP, phpMyAdmin

Platforms & Development: Windows, macOS, Linux, GCP, Netlify, Supabase, Godot